07/01/2026

Code-B

Aakash
Medical|IIT-JEE|Foundations

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MM : 720 NCERT Booster Test Series for NEET-2026_RM(P2)_NBTS-03B **Time : 180 Min.**

Topics Covered:
Physics: Electrostatic Potential and Capacitance, Current Electricity, Moving Charges and Magnetism, Magnetism and Matter, Electromagnetic Induction, Alternating Current
Chemistry: Chemical Kinetics, Organic Chemistry: Some Basic Principles & Techniques, Hydrocarbons, Haloalkanes and Haloarenes
Botany: Photosynthesis in Higher Plants, Respiration in Plants, Plant Growth and Development, Sexual Reproduction in Flowering Plants
Zoology: Animal Kingdom, Structural Organisation in Animals–Animal Morphology (Cockroach & Frog)

General Instructions :
Duration of Test is 3 hrs.
The Test consists of 180 questions. The maximum marks are 720.
There are four parts in the question paper consisting of Physics, Chemistry, Botany and Zoology having 45 questions in each part of equal weightage.
Each question carries +4 marks. For every wrong response, –1 mark shall be deducted from the total score. Unanswered/unattempted questions will be given no marks.
Use blue/black ballpoint pen only to darken the appropriate circle.
Mark should be dark and completely fill the circle.
Dark only one circle for each entry.
Dark the circle in the space provided only.
Rough work must not be done on the Answer sheet and do not use white fluid or any other rubbing material on the Answer sheet.

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PHYSICS

1. Consider a series LCR circuit in which $L = \frac{2}{\pi}$ H, $C = \frac{100}{\pi}\mu$F and resistance R = 100 Ω. When circuit is connected with an a.c. source 220 V, 50 Hz. The current drawn from the source will be

   (1) $\sqrt{2}$ A

   (2) $\frac{11\sqrt{2}}{10}$ A

   (3) $\sqrt{5}$ A

   (4) $\frac{11\sqrt{5}}{10}$ A

2. In a certain region potential is given by $V = k(2xy)$. The electric field at (1, 1) has magnitude

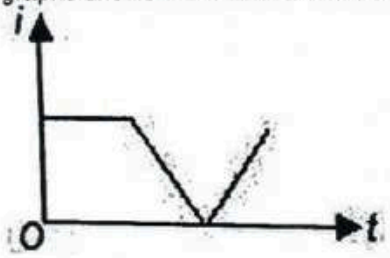
   (1) $\sqrt{2}\,k$

   (2) $2\sqrt{2}\,k$

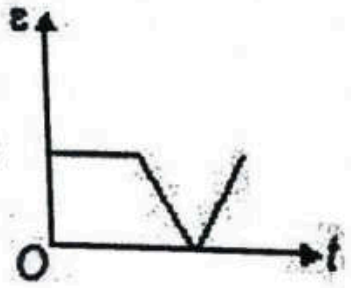
   (3) $\frac{k}{\sqrt{2}}$

   (4) k

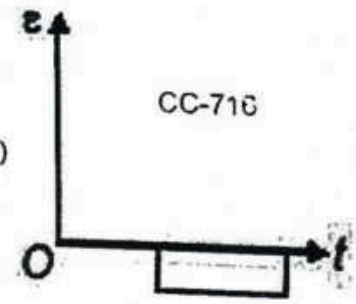
3. The current i in an induction coil varies with time t according to the graph shown in the figure. Which of the following graphs shows the induced emf ε in the coil with time



(1)

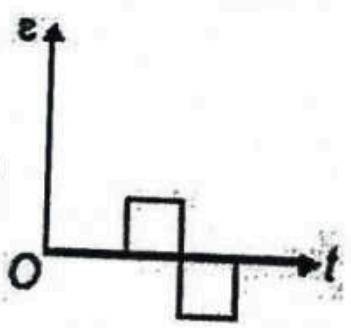


(2)

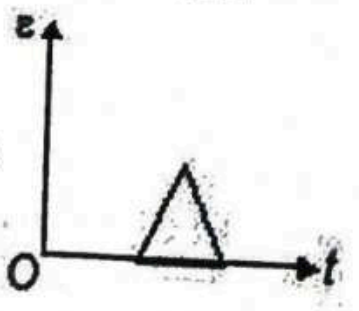


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(3)



(4)



4. Two conducting wires A and B of same dimensions have electron density ratio 1 : 3. If their relaxation time is same, then the ratio of resistance of A to resistance of B is

CC-716

(1) 1 : 1

(2) 3 : 1

(3) 1 : 3

(4) 9 : 1

5. A metallic rod of mass per unit length 0.5 kg m^{-1} is lying horizontally on a smooth inclined plane which makes an angle of 30° with the horizontal. The rod is not allowed to slide down by flowing a current through it when a magnetic field of induction 0.25 T is acting on it in the vertical direction. The current flowing in the rod to keep it stationary is

(1) 7.14 A

(2) 5.98 A

(3) 11.32 A

(4) 14.76 A

6. The coercivity of a bar magnet is 100 A/m. It is to be demagnetised by placing it inside a solenoid of length 100 cm and number of turns 50. The current flowing in the solenoid will be

(1) 4 A

(2) 2 A

(3) 1 A

(4) Zero

CC-716

7. A direct current of 2 A and an alternating current having a maximum value of 2 A flow through two identical resistance. The ratio of heat produced in the two resistance will be

CC-716

(1) 2 : 1

(2) 1 : 1

(3) 4 : 1

(4) 1 : 8

8. The voltage of an AC source varies with time according to the relation $E = 200 \sin(100\pi t) \cos(100\pi t)$ V, where t is time in second.

Select **correct** statement.

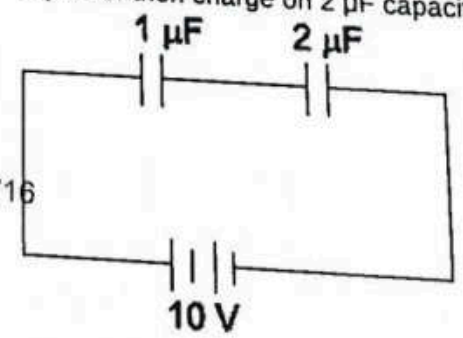
(1) The peak voltage of source is 200 V

(2) The peak voltage of source is 400 V

(3) The frequency of source voltage is 50 Hz

(4) The frequency of source voltage is 100 Hz

9. As shown in figure, if a dielectric $K = 2$ is inserted into 1 μF capacitor then charge on 2 μF capacitor will



CC-716

CC-716

(1) Increase

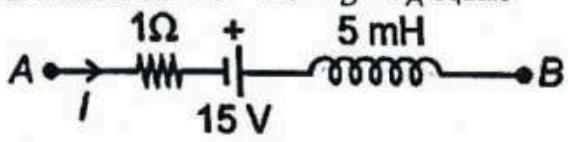
(2) Decrease

(3) Remain same

(4) Need more information

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10. The network shown in the figure is part of a complete circuit. If at a certain instant, the current I is 5 A and it is decreasing at a rate of 10^3 As^{-1} then $V_B - V_A$ equals



(1) 20 V

(2) 15 V

(3) 10 V

(4) 5 V

11. Which of the following statements regarding magnetic lines of force is correct?

(1) Total magnetic flux linked with a closed surface is always zero

(2) They need not be perpendicular to the surface from where they start or where they meet

(3) They may or may not pass through a conductor

(4) All of these

12. **Assertion (A) :** A bar magnet creates non-uniform magnetic field.
Reason (R) : Net magnetic force on a bar magnet placed in an uniform magnetic field is zero.

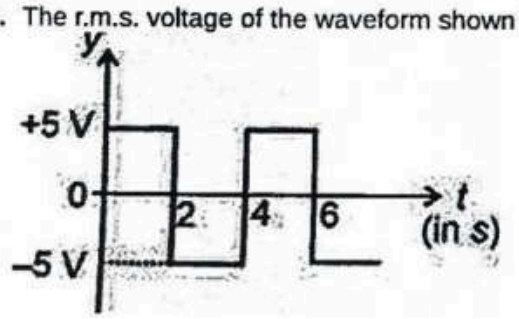
(1) Both Assertion & Reason are true and the reason is the correct explanation of the assertion.

(2) Both Assertion & Reason are true but the reason is not the correct explanation of the assertion.

(3) Assertion is true statement but Reason is false.

(4) Both Assertion and Reason are false statements.

13. The r.m.s. voltage of the waveform shown is



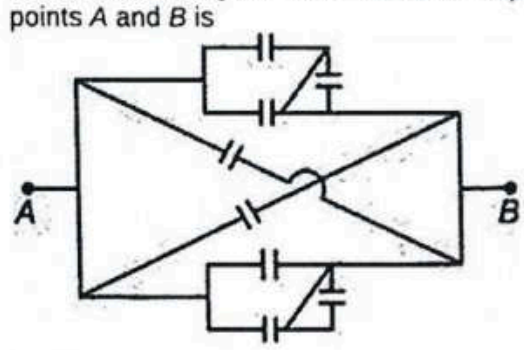
(1) 5 V

(2) 10 V

(3) 15 V

(4) 20 V

14. Eight capacitors, each of capacitance 2 μF are connected as shown in the figure. The effective capacitance between points A and B is



(1) 10 μF

(2) 12 μF

(3) 8 μF

(4) Zero

15. Match column-I with column-II.

Column-I (Material)		**Column-II (Relative permeability μ_r)**
a. Diamagnetic	(i)	$1 < \mu_r < 1 + \varepsilon$, where ε is a small and positive number
b. Ferromagnetic	(ii)	$0 \le \mu_r < 1$
c. Paramagnetic	(iii)	$\mu_r >> 1$

(1) a(iii), b(ii), c(i)

(2) a(ii), b(i), c(iii)

(3) a(i), b(iii), c(ii)

(4) a(i), b(ii), c(iii)

16. **Assertion (A):** The conductivity of conductor increases with increase in temperature.
Reason (R): With increase in temperature, drift speed of free electrons inside conductor increases.
In the light of above statements, choose the correct option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

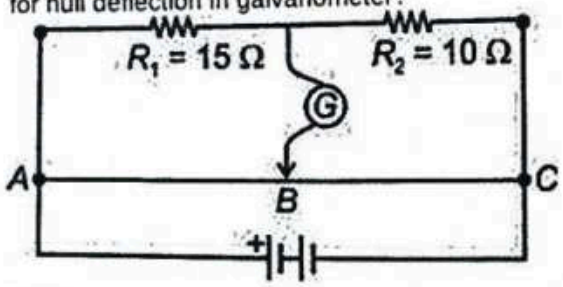
(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

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17. In the meter bridge circuit shown in figure, what is length *AB* for null deflection in galvanometer?



(1) 40 cm

(2) 50 cm

(3) 60 cm

(4) 70 cm

18. A short bar magnet of magnetic moment 0.4 J/T is placed in a uniform magnetic field of 0.24T. The magnet is in stable equilibrium when the potential energy is

(1) 0.006 J

(2) 0.001 J

(3) –0.096 J

(4) –0.048 J

19. The flux through a circular conducting coil of resistance 20 Ω is varying with time (t) as $\phi = 5t^2$ Wb. The magnitude of induced current in the loop at time t = 1s is

(1) 0.5 A

(2) 1 A

(3) 0.25 A

(4) 10 A

20. The magnetic flux linked through a plane having area A placed in uniform magnetic field B will be maximum, if angle between the area vector $\vec{A}$ and magnetic field $\vec{B}$ will be

(1) 45°

(2) 90°

(3) Zero

(4) 60°

21. A capacitor of 20 μF is completely charged with the battery of 100 V. Now battery is removed and this capacitor is connected with another identical uncharged capacitor. The common potential of the combination will be

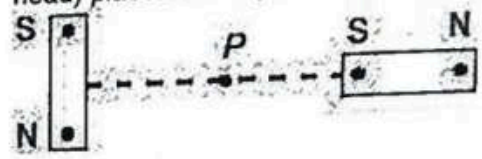
(1) 200 V

(2) 50 V

(3) 100 V

(4) 25 V

22. Two identical bar magnets are kept as shown in the figure. The direction of a magnetic compass, (indicated by arrow head) placed at the point P is (approximately)



(1) →

(2) ↗

(3) ↘

(4) ↑

23. Given below are two statements:
Statement I : The 220 V of domestic AC circuit represents peak voltage.
Statement II : Average value of sinusoidal AC current may have zero value in half time period. (where symbols have their usual meaning).
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both Statement I and Statement II are correct

(2) Both Statement I and Statement II are incorrect

(3) Statement I is correct but Statement II is incorrect

(4) Statement I is incorrect but Statement II is correct

24. The correct statement among the following is

(1) Voltmeter has high resistance and is placed in parallel across the load resistance in a circuit

(2) Ammeter has low resistance and is placed in series with the load resistance in a circuit

(3) Current sensitivity of galvanometer is current per unit deflection

(4) Both (1) and (2)

25. The space between the plates of a parallel plate capacitor is filled with dielectric, while the plates remain connected to the battery. Then
a. Energy stored in it increases.
b. Energy stored in it decreases.
c. Its capacitance increases.
d. The product of charge and voltage increases.
e. The ratio of charge to voltage decreases.
The correct statement(s) is/are

(1) b, c and d only

(2) a, c and d only

(3) a, c and e only

(4) a, c, d and e only

26. A proton and an α-particle, having same kinetic energies are moving in a circular path in uniform magnetic field B. Then

(1) The radius of circular path is same for both

(2) Both will have different radius of circular path

(3) Both have same time period of their circular motion

(4) None of these

CC-716 CC-716 CC-716 CC-716 CC-716

27. Two long parallel straight wires A and B, separated by a distance of 6 cm, carries currents I and $2I$ respectively in the same direction. The distance from wire B at which the net field will be zero, is

(1) 1 cm

(2) $\frac{3}{2}$ cm

(3) 4 cm

(4) 5 cm

28. Read the following Assertion (A) and Reason (R) statements and choose the correct option.
A : Reactance offered by a capacitor increases with the increase in frequency of the AC source.
R : The current leads the voltage in purely capacitive network by $\frac{\pi}{2}$.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

29. A series LCR circuit is connected across a source of emf $E = 20\sin\left(50\pi t - \frac{\pi}{3}\right)$ V. If a current $I = 10\sin 50\pi t$ A flows in the circuit, then the power dissipated through circuit is (t is in s)

(1) $50\sqrt{2}$ W

(2) 50 W

(3) $50\sqrt{3}$ W

(4) 100 W

30. When a 20 V m^{-1} electric field is produced inside a conductor and 4×10^4 A m^{-2} current density is established in it. Resistivity of the conductor is

(1) $5 \times 10^{-4}\ \Omega$ m

(2) $4 \times 10^{-3}\ \Omega$ m

(3) $2 \times 10^{-2}\ \Omega$ m

(4) $3 \times 10^{-4}\ \Omega$ m

31. Which of the following statement is correct about induced electric field?

(1) This field is produced by variation of magnetic field with respect to space.

(2) This field is conservative in nature.

(3) We cannot define a potential corresponding to this field.

(4) The lines of induced electric field are not closed curve.

32. A magnetic needle of magnetic moment $1280\pi^2 \times 10^{-6}$ A m^2 is placed in uniform magnetic field of 0.049 T. If moment of inertia of the needle is 9.8×10^{-6} kg m^2, then time period of oscillation of magnetic needle in the magnetic field is

(1) $\frac{1}{2}$ s

(2) 1 s

(3) $\frac{1}{4}$ s

(4) $\frac{1}{8}$ s

33. An electric dipole of dipole moment $\left(\vec{P}\right)$ is placed in a uniform electric field $\left(\vec{E}\right)$ in the direction of electric field. The work done by external force to rotate it by angle θ is $\frac{PE}{2}$, then value of θ will be

(1) 90°

(2) 60°

(3) 120°

(4) 180°

34. An angle $\angle CAD$ made of conducting wire moves along its bisector through a magnetic field B as suggested by figure. The emf induced between the ends C and D if the magnetic field is perpendicular to the plane of the wire, is [CA = AD = l



(1) $\sqrt{2}Blv$

(2) $\sqrt{3}Blv$

(3) $\frac{\sqrt{3}}{2}Blv$

(4) $\frac{1}{2}Blv$

35. A solenoid of length 0.5 m has a radius of 1 cm and is made up of 500 turns. If the magnetic field inside the solenoid is 6.28×10^{-3} T, what is the current flowing through the solenoid?

(1) 10 A

(2) 5 A

(3) 15 A

(4) 20 A

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36. Keeping volume constant of the conductor, if the length of conductor is made n-times, then resistance of the conductor becomes

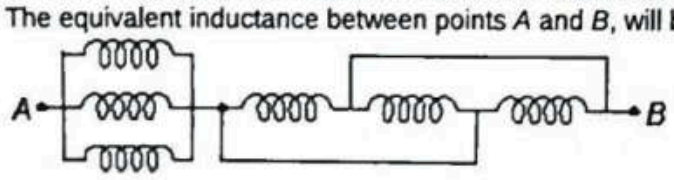
(1) n^2 times its initial value

(2) n^3 times its initial value

(3) $\frac{1}{n^2}$ times its initial value

(4) $\frac{1}{n}$th times its initial value

37. All the inductors shown below have same inductance of 6 H. The equivalent inductance between points A and B, will be



(1) 1 H

(2) 3 H

(3) 2 H

(4) 4 H

38. The power supplied by the magnetic force acting on a moving charged particle

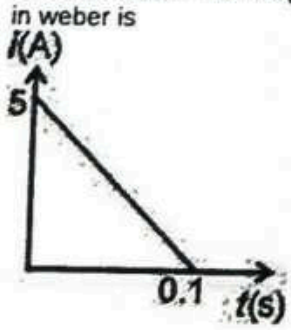
(1) Is never zero

(2) Is always zero

(3) Is zero only when $\vec{v}$ is perpendicular to $\vec{B}$

(4) Is zero only when $\vec{v}$ is parallel to $\vec{B}$

39. Magnetic flux changes in a coil of resistance 10 Ω. As a result an induced current is developed in it, which varies with time as shown. The magnitude of change in flux through coil in weber is



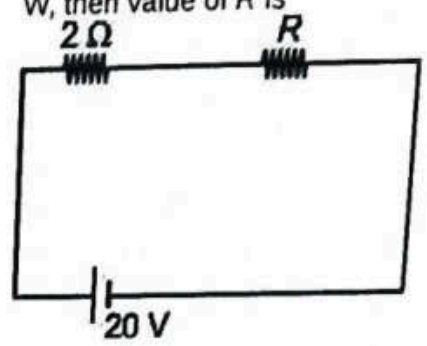
(1) 2

(2) 2.5

(3) 4

(4) 6

40. If the power dissipation in the circuit shown in figure is 100 W, then value of R is



(1) 1 Ω

(2) 2 Ω

(3) 3 Ω

(4) 4 Ω

41. A magnet of magnetic moment $\vec{M} = \left(50\hat{i}\right) A\,m^2$ is placed in a magnetic field $\vec{B} = \left(0.2\hat{i} + 0.3\hat{j}\right)$ T. The torque acting on the magnet is

CC-716

(1) $\left(15\hat{k}\right)$ N m

(2) $\left(15\hat{k} + 10\hat{j}\right) N\,m$

(3) $10\hat{k}$ N m

(4) Zero

42. Consider the following two statements and choose the correct option.

Statement-I: Electric field inside conductor is always zero.

Statement-II: Under electrostatic condition electric field inside conductor is non-zero.

(1) Only statement I is true

(2) Only statement II is true

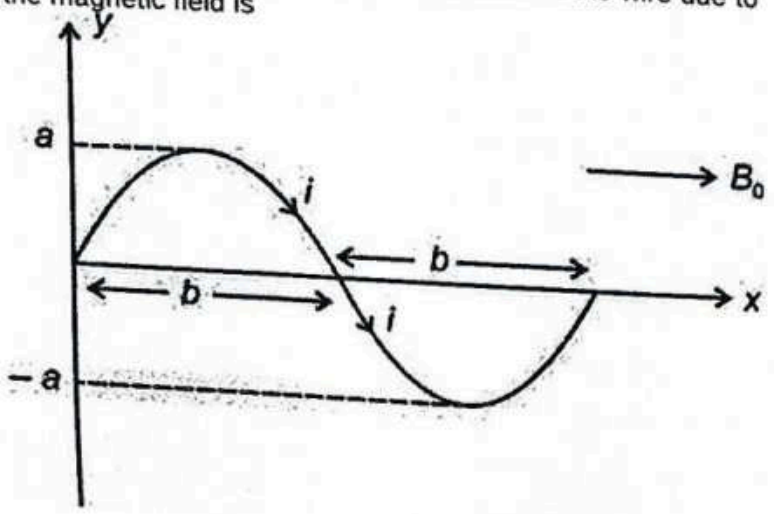
(3) Both statement I and statement II are true

(4) Neither statement I nor statement II are true

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CC-716 CC-716

43. A conducting wire in the form of sine curve is placed in x-y plane in a uniform magnetic field directed along positive x-axis as shown in the figure. The net force on the wire due to the magnetic field is



(1) $2bB_0i$

(2) bB_0i

(3) $2aB_0i$

(4) Zero

44. Choose the wrong statement.

(1) Potential difference across the terminals of a cell in a closed circuit is always more than its EMF during discharging.

(2) Internal resistance of an ideal cell is zero.

(3) In discharging, potential difference versus current graph for a cell is a straight line with a -ve slope.

(4) Terminal potential difference of the cell when it is being charged is given as $V = E + Ir$.

45. A cell of emf E and internal resistance r gives a current of 0.5 A with an external resistance of 12 Ω and a current of 0.25 A with an external resistance of 25 Ω. What is the value of the internal resistance of the cell?

(1) 5 Ω

(2) 1 Ω

(3) 7 Ω

(4) 3 Ω

CHEMISTRY

46. Select an incorrect statement about a catalyst

(1) It does not affect the energy of activation

(2) It increases the rate of reaction

(3) It does not alter ΔG of a reaction

(4) It does not change the equilibrium constant of a reaction

47. For a zero order reaction, the slope of plot $[A_t]$ vs t is

(1) $-\frac{0.693}{k}$

(2) $-\frac{1}{k}$

(3) $-k$

(4) $\frac{1}{k}$

48. If the rate constant for a first order reaction is k then the time required for 99.99% completion of the reaction is

(1) $\frac{2.303}{k}$

(2) $\frac{3\times2.303}{k}$

(3) $\frac{4\times2.303}{k}$

(4) $\frac{10\times2.303}{k}$

CC-716 CC-716 CC-716 CC-716 CC-716

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49. Given below are two statements:
Statement I: The decomposition of gaseous ammonia on a hot platinum surface is a zero-order reaction at high pressure.
Statement II: $t_{1/2}$ for a zero order reaction is directly proportional to the initial concentration of the reactants.
In the light of the above statements, choose the most appropriate answer from the options given below.

(1) Both statement I and statement II are correct

(2) Both statement I and statement II are incorrect

(3) Statement I is correct but statement II is incorrect

(4) Statement I is incorrect but statement II is correct

50. In the reaction $2NH_3 \rightarrow N_2 + 3H_2$.

If the rate of reaction is 0.6 $molL^{-1}s^{-1}$. The rate of production of H_2 is

(1) 1.8 $molL^{-1}s^{-1}$

(2) 0.9 $molL^{-1}s^{-1}$

(3) 0.2 $molL^{-1}s^{-1}$

(4) 1.2 $molL^{-1}s^{-1}$

51. Mechanism of a hypothetical reaction,
$2A_3 \rightarrow 3A_2$
is given below as

(i) $A_3 \rightleftharpoons A_2 + A$ (fast)

(ii) $A_3 + A \rightarrow 2A_2$ (slow)

Rate law for the above reaction will be

(1) $r = K[A_3]^2[A_2]^{-1}$

(2) $r = K[A_3][A_2]$

(3) $r = K[A_3]^2[A_2]$

(4) $r = K[A_3]^2$

52. For the reaction A + B → C + D, on doubling the concentration of both the reactants A and B, the reaction rate becomes 4 times and on doubling the concentration of only B, the rate of reaction remains the same. The rate law for the above reaction is

(1) $r = k[A]^1[B]^1$

(2) $r = k[A]^0[B]^1$

(3) $r = k[A]^2[B]^0$

(4) $r = k[A]^0[B]^2$

53. Consider the following statements regarding to the equation,
Rate = $PZ_{AB}e^{-E_a/RT}$
(a) Z_{AB} represents the collision frequency of reactants, A and B.
(b) $e^{-E_a/RT}$ represents the fraction of molecules with energies less than E_a.
(c) P takes into account the fact that in a collision, molecules must be properly oriented.
Select the correct statements.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

54. For a chemical reaction rate of reaction doubles on increasing temperature from 300 K to 310 K, then activation energy for the reaction will be

(1) 18.4 kJ

(2) 106.8 kJ

(3) 24.44 kJ

(4) 53.42 KJ

55. Match order of reaction in list-I with units of rate constant in list-II

	List-I		List-II
a.	Zero order	(i)	$Mol^{-1}\,L\,s^{-1}$
b.	First order	(ii)	$mol\,L^{-1}\,s^{-1}$
c.	Second order	(iii)	$mol^{-2}\,L^2\,s^{-1}$
d.	Third order	(iv)	s^{-1}

Correct match is

(1) a(iii), b(i), c(ii), d(iv)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(ii), b(iv), c(i), d(iii)

(4) a(ii), b(iv), c(iii), d(i)

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56. Match the **List-I** of the given process with their respective name reaction given in **List-II**

List-I	List-II
a. Conversion of aryl halide into diphenyl	(i) Finkelstein reaction
b. Conversion of alkyl chloride into alkyl fluoride	(ii) Sandmeyer reaction
c. Conversion of benzene diazonium halide into aryl halide	(iii) Swarts reaction
d. Conversion of alkyl bromide into alkyl iodide	(iv) Fittig reaction

Select the **correct** option.

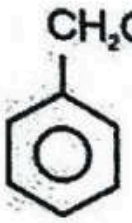
(1) a(ii), b(iv), c(i), d(iii)

(2) a(ii), b(i), c(iv), d(iii)

(3) a(iv), b(i), c(iii), d(ii)

(4) a(iv), b(iii), c(ii), d(i)

57. Which of the following will give fastest S_N1 reaction?

(1) 

CC-716

(2) $CH_2 = CH - CH_2Cl$

(3) $CH_3-\underset{\underset{Cl}{|}}{CH}-CH_3$

(4) $CH_3-\underset{\underset{CH_3}{|}}{\overset{\overset{Cl}{|}}{C}}-CH_3$

58. Given below are the two statements: one is labelled as Assertion (A) and the other is labelled as Reason (R).

Assertion (A) : Haloalkanes react with AgCN to form alkyl cyanide as the major product.

Reason (R) : AgCN is mainly covalent in nature and nitrogen is free to donate electron pair forming isocyanide as the major product.

In the light of above statement, choose the **correct** option.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

CC-716

(3) (A) is true but (R) is false

(4) (A) is false but (R) is true

59.

$$CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_3 + Cl_2 \xrightarrow{h\nu} \underset{(1\ eq)}{}$$

Major product for above reaction will be

(1) $CH_3-\overset{\overset{Cl}{|}}{\underset{\underset{CH_3}{|}}{C}}-CH_3$

(2) $CH_3-\underset{\underset{CH_3}{|}}{CH}-CH_2-Cl$

(3) $CH_3-CH_2-CH_2-CH_2-Cl$

(4) $CH_3-\underset{\underset{Cl}{|}}{CH}-CH_2-CH_3$

60. Consider the following statements

CC-716 (a) In S_N1 reaction, carbocation intermediate is formed CC-716

(b) S_N1 reactions are generally carried out in polar protic solvents

(c) Greater the stability of carbocation faster is S_N1 rate of reaction

The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

61. Given below are two statements one is labelled as Assertion (A) and other is labelled as Reason (R)

Assertion (A) : Boiling points of methyl halides decreases in the order : $CH_3I > CH_3Br > CH_3Cl > CH_3F$

Reason (R) : With the increase in size and mass of halogen atom, the magnitude of van der Waal forces increases.

In the light of above statements, choose the most appropriate answer from the options given below.

(1) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(2) Both (A) and (R) are true and (R) is the correct explanation of (A)

(3) Both (A) and (R) are false

CC-716 (4) (A) is true but (R) is false CC-716

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62. Consider the following statements.
Statement (I) : Haloarenes are less reactive than haloalkanes towards nucleophilic substitution reaction.
Statement (II) : In haloarenes, the electron pairs on halogen atom are in conjugation with π-electrons of the ring.
In the light of above statements, choose the correct option.

(1) Statement (I) is correct but Statement (II) is incorrect

(2) Statement (I) is incorrect but Statement (II) is correct

(3) Both Statement (I) and Statement (II) are correct

(4) Both Statement (I) and Statement (II) are incorrect

63. Chloroform is stored in closed dark coloured bottles completely filled to avoid its slow oxidation in the presence of light and air. The extremely poisonous gas formed during this reaction is

(1) Acetylene

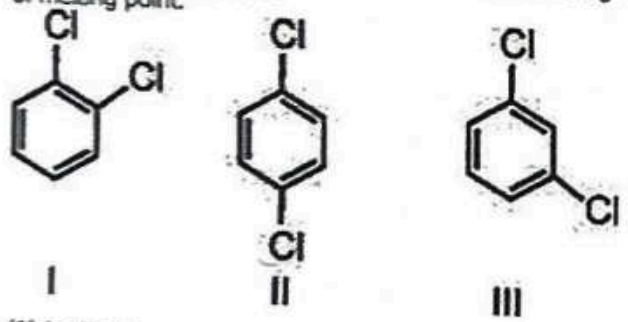
(2) Chlorine

(3) Carbon monoxide

(4) Phosgene

64. Arrange the following compounds in their decreasing order of melting point.

CC-716



I II III

(1) I > II > III

(2) III > I > II

(3) II > I > III

(4) II > III > I

65. When ethyl bromide is treated with sodium in presence of dry ether it yields

(1) But-2-ene

(2) Ethane

(3) Ethene

(4) Butane

66. Given below are two statements.
Statement I: Thin layer chromatography is a type of adsorption chromatography.
Statement II: Paper chromatography is a type of partition chromatography.
In the light of above statements, choose the correct option.

(1) Statement I is false but Statement II is true

(2) Both statement I and statement II are true

(3) Both statement I and statement II are false

(4) Statement I is true but statement II is false

67. Increasing order of priority of functional group according to the IUPAC nomenclature is

(1) $-COOH < -OH < -NH_2 < -COCl$

(2) $-NH_2 < -OH < -COOH < -COCl$

(3) $-COOH < -COCl < -OH < -NH_2$

(4) $-NH_2 < -OH < -COCl < -COOH$

68. In Carius method of estimation of Bromine 1 g of an organic compound gave 1.128 g of AgBr. The percentage of bromine in the compound is
[Molar mass of AgBr = 188 g mol^{-1}, gram atomic mass of Br = 80 g mol^{-1}]

(1) 75%

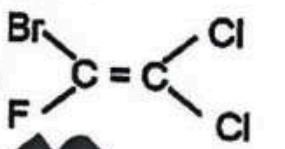
(2) 48%

(3) 35%

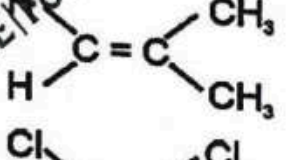
(4) 26%

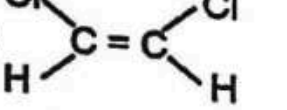
69. Which of the following compounds can show geometrical isomerism?

CC-716

(1) 

(2) 

(3) 

(4) 

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70. In sulphur estimation, 0.16 g of an organic compound gave 0.466 g of barium sulphate. The percentage of sulphur in the organic compound is

(1) 31.91%

(2) 40%

(3) 35.76%

(4) 20.27%

71. Functional isomer of carboxylic acid is

(1) Aldehyde

(2) Ketone

(3) Ester

(4) Anhydride

72. Which compound on reaction with sodium, liberates hydrogen gas?

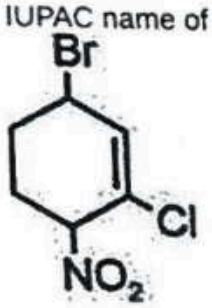
(1) $CH_2=CH_2$

(2) $CH_3–C\equiv C–CH_3$

(3) $CH_3–C\equiv CH$

(4) $CH_3–CH_2–CH_2–CH_2–CH_3$

73. IUPAC name of the following compound is



(1) 1-Bromo-3-chloro-4-nitrocyclohex-2-ene

(2) 5-Bromo-1-chloro-6-nitrocyclohex-5-ene

(3) 3-Bromo-1-chloro-6-nitrocyclohex-1-ene

(4) 4-Bromo-2-chloro-1-nitrocyclohex-2-ene

CC-716

74. In the Kjeldahl's method for estimation of nitrogen present in an organic sample, ammonia evolved from 0.80 g of sample is neutralised by 20 ml of 0.6 M H_2SO_4. The percentage of nitrogen in the organic sample is

(1) 50%

(2) 42%

(3) 48%

(4) 52%

75. Arrange the given carbonium ions in increasing order of stability



(1) (iii) > (ii) > (i)

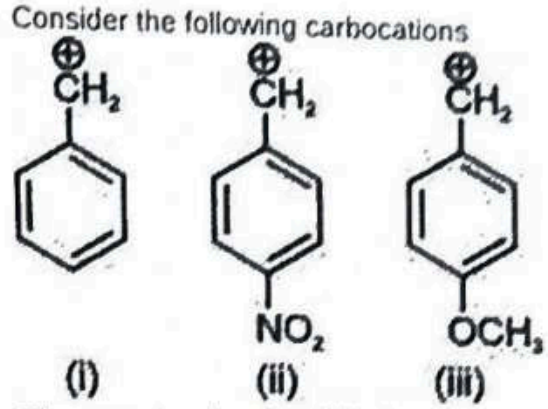
(2) (ii) > (iii) > (i)

(3) (i) > (iii) > (ii)

(4) (ii) > (i) > (iii)

CC-716

76. Consider the following carbocations



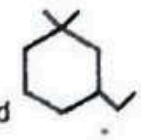
The correct order of stability is

(1) (ii) > (i) > (iii)

(2) (ii) > (iii) > (i)

(3) (iii) > (ii) > (i)

(4) (iii) > (i) > (ii)

77. IUPAC name of the compound  is

CC-716

(1) 3-ethyl-1,1-dimethylcyclohexane

(2) 1-ethyl-3,3-dimethylcyclohexane

(3) (3,3-dimethylcyclohexyl) ethane

(4) 1,3-diethylcyclohexane

78. All of the following species are nucleophiles, **except**

(1) CH_3OH

(2) CH_3NH_2

(3) $AlCl_3$

(4) NH_3

79. Choose the **incorrect** statement.

(1) Cis-but-2-ene is more polar than trans-but-2-ene

(2) Dipole moment of trans-but-2-ene is almost zero

(3) Boiling point of trans-but-2-ene is higher than that of cis-but-2-ene

(4) It is observed that the trans but-2-ene has higher melting point than cis-but-2-ene

80. The correct increasing order of boiling points of the following alkanes is

a. n-Hexane

b. n-Pentane

c. 2-methylbutane

d. 2,3-dimethylpropane

Choose the **correct** answer from the following options.

CC-716

(1) a < b < c < d

(2) d < b < c < a

(3) d < c < b < a

(4) b < d < c < a

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81. Consider the following statements.
(a) There are infinite number of conformations of ethane.
(b) The repulsive interaction between the electron clouds, which affects stability of a conformation, is called torsional strain.
(c) In ethane, magnitude of torsional strain does not depend upon the angle of rotation about C – C bond.
The correct statements are

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) (a), (b) and (c)

82. An alkene on vigorous oxidation with hot acidified $KMnO_4$ gives only acetic acid, the alkene is

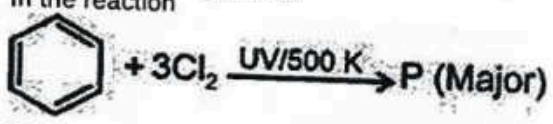
(1) $CH_3CH_2CH = CH_2$

(2) $CH_3CH = CHCH_3$

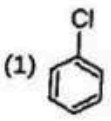
(3) $(CH_3)_2C = CH_2$

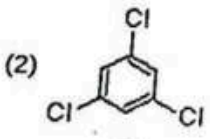
(4) $CH_3CH_2CH = CHCH_3$

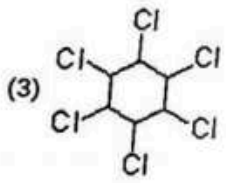
83. In the reaction CC-716

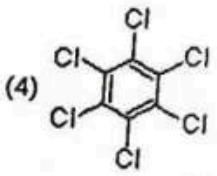


P is

(1) 

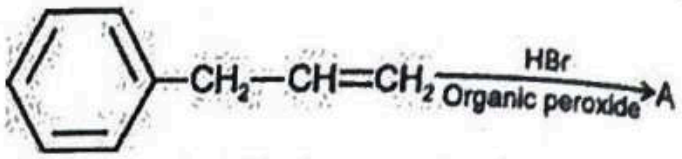
(2) 

(3) 

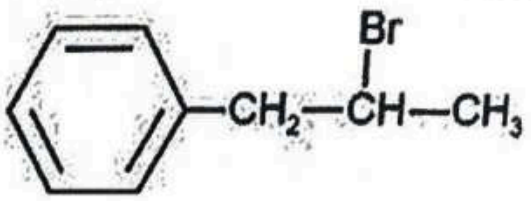
(4) 

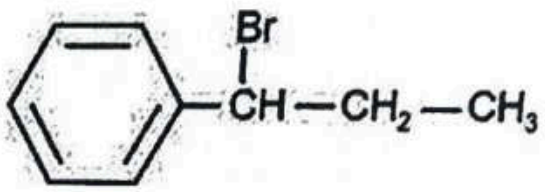
CC-716

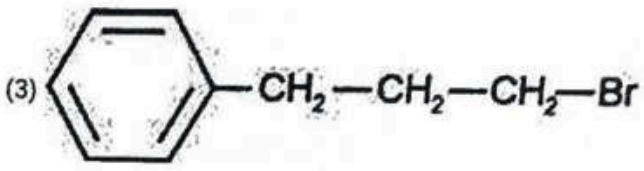
84.



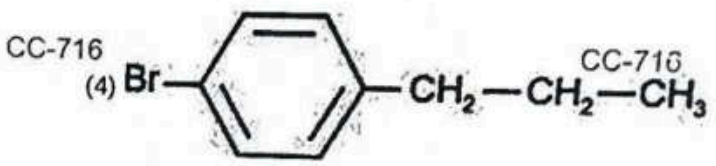
The major product A in above mentioned reaction is

(1) 

(2) 

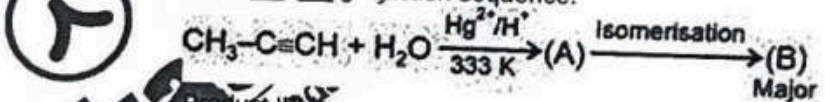
(3) 

CC-716

(4) 

CC-716

85. In the following reaction sequence.



Product is

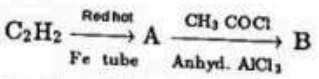
(1) Propan-2-ol

(2) Propan-1-ol

(3) Propanal

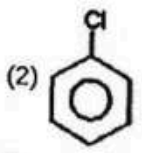
(4) Propanone

86.

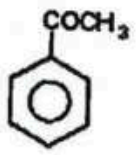


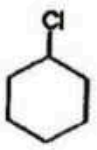
Product B is

(1) C_2H_6

(2) 

CC-716

(3) 

(4) 

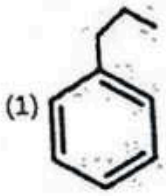
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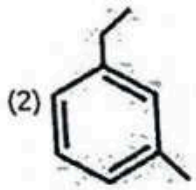
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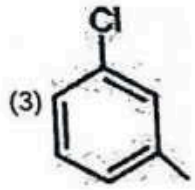
87. In the following reaction sequence

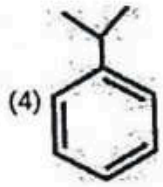
$$CH_3-(CH_2)_4-CH_3 \xrightarrow[773\ K,\ 10\text{–}20\ atm]{Mo_2O_3} \underset{Major}{(A)} \xrightarrow[Anhyd.\ AlCl_3]{CH_3CH_2CH_2Cl} \underset{Major}{(B)}$$

The product (B) is

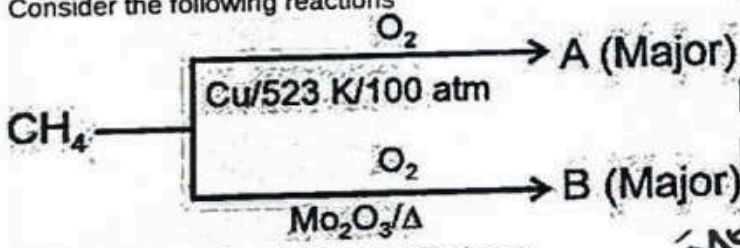
(1) 

(2) 

(3) 

(4) 

88. Consider the following reactions



Major products (A) and (B) respectively are

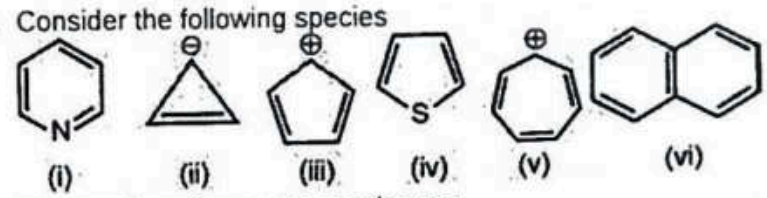
(1) HCOOH and HCHO

(2) CH_3OH and HCHO

(3) CH_3OH and HCOOH

(4) HCHO and CH_3OH

89. Consider the following species



(i) (ii) (iii) (iv) (v) (vi)

The number of aromatic species are

(1) 2

(2) 3

(3) 4

(4) 5

90. In Friedel craft acylation, the electrophile is

(1) $CH_3\overset{\oplus}{C}H_2$

(2) $\overset{\oplus}{Cl}$

(3) $\overset{\oplus}{CH_3}$

(4) $CH_3\overset{O}{\overset{\|}{C}}\oplus$

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BOTANY

91. The stromal lamella of chloroplast connects

(1) Two adjoining grana

(2) Two adjoining starch granules

(3) Granum with inner membrane

(4) Lipid droplet with starch granule

92. Which of the following statements is **not** true?

(1) The NADP reductase enzyme is located on the stroma side of the thylakoid membrane

(2) During reduction of $NADP^+$ to NADPH + H^+, the protons are removed from the stroma

(3) Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATP synthase

(4) A conformational change in the reaction centre of the photosystem, results in synthesis of ATP molecules

93. Read the statements given below and choose the **correct** option.

Statement A: T.W. Engelmann used a prism to split light into its spectral components and illuminated *Cladophora* placed in a suspension of aerobic bacteria to detect the site of O_2 evolution.

Statement B: Jan Ingenhousz conducted the bell jar experiment to prove that plants restore to the air whatever breathing animals and burning candle remove.

(1) Both the statements A and B are incorrect

(2) Both the statements A and B are correct

(3) Only statement A is correct

(4) Only statement B is correct

94. "The productivity and yields are better in C_4 plants as compared to C_3 plants". This is due to

(1) Presence of thin cuticle in C_4 plants

(2) Higher surface area of leaves in C_4 plant

(3) Higher water use by C_4 plant

(4) Presence of a mechanism in C_4 plants that minimizes the oxygenase activity of RuBisCO and increases the carboxylase activity

95. Mark the **incorrect** statement(s) w.r.t Z-scheme.

(a) Only PS-II is involved

(b) Only PS-I is involved

(c) Splitting of H_2O is associated with PS-I

(d) In this scheme of light reaction electrons do not move downhill on redox potential scale

(1) Only (a), (b) and (c)

(2) Only (a), (b) and (d)

(3) (a), (b), (c) and (d)

(4) Only (d)

96. Match List-I with List-II and select the **correct** option.

	List-I		List-II
a.	Chlorophyll *b*	(i)	Protects the chief pigment from photo-oxidation
b.	Xanthophyll	(ii)	It is the chief photosynthetic pigment
c.	Chlorophyll *a*	(iii)	It appears yellow in the chromatogram
d.	Carotenoid	(iv)	It appears yellow-green in chromatogram

(1) a(iv), b(i), c(iii), d(ii)

(2) a(iv), b(iii), c(ii), d(i)

(3) a(iii), b(iv), c(ii), d(i)

(4) a(ii), b(i), c(iv), d(iii)

97. The photorespiration pathway

(1) Does not occur in C_4-plants

(2) Does not occur in C_3-plants

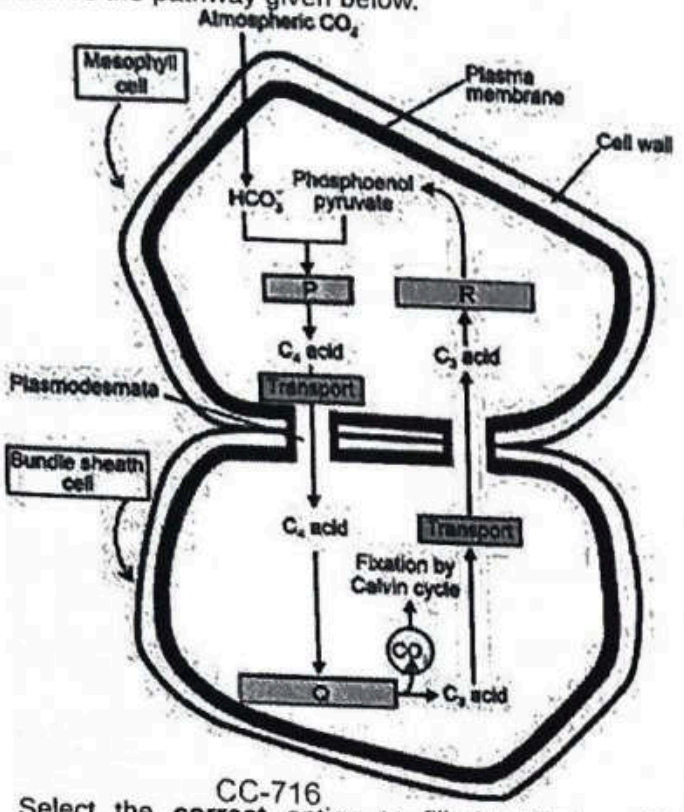
(3) Produces many ATP molecules

(4) Is mainly involved in sugar synthesis

CC-716

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98. Observe the pathway given below.



CC-716

Select the **correct** option to fill the blanks, which are indicated by P, Q and R.

	P	Q	R
(1)	Reduction	Carboxylation	Regeneration
(2)	Fixation	Regeneration	Decarboxylation
(3)	Carboxylation	Regeneration	Decarboxylation
(4)	Fixation	Decarboxylation	Regeneration

(1) (1)

(2) (2)

(3) (3)

(4) (4)

99. Photosynthesis is divided into the light dependent reaction and Dark reaction.
Regarding these processes, which of the given statements is **true**?

(1) During the light dependent reaction, there is fixation of CO_2 to produce glucose

(2) The Calvin cycle involves conversion of water molecules into oxygen as a byproduct of dark reaction

(3) The dark reaction involves several protein complexes and pigments

(4) The light dependent reaction occurs in thylakoid membrane and the Calvin cycle in the stroma region of chloroplast

100. ATP synthesis in chloroplast takes place

(1) In the intermembrane space

(2) When there is movement of protons from stroma to thylakoid lumen

(3) Towards the stroma side of thylakoid membrane

(4) Towards the cytoplasm through outer membrane

101. Accessory pigments

(1) Absorb light of different wavelengths and prevent photo-oxidation of chlorophyll

(2) Include all pigments other than chlorophyll b

(3) Are called antennae molecule which take energy from chlorophyll molecule

(4) Can absorb only narrow range of wavelength of visible spectrum

102. How many ATP and NADPH are required to form five molecules of glucose in tomato plant?

(1) 90 NADPH and 60 ATP

CC-716 (2) 90 ATP and 90 NADPH CC-716

(3) 60 NADPH and 90 ATP

(4) 150 ATP and 60 NADPH

103. The complete oxidation of pyruvate by the stepwise removal of all the hydrogen atoms, leaving three molecules of CO_2 occurs in the

(1) Inner membrane of thylakoid

(2) Matrix of the mitochondria

(3) Outer membrane of mitochondria

(4) Inter-membrane space of chloroplast

104. Read the following statements and identify the **correct** ones w.r.t. ETC.
a. Electrons from NADH produced in the mitochondrial matrix during TCA cycle are oxidised by NADH dehydrogenase
b. Ubiquinone receives reducing equivalents via $FADH_2$
c. The ubiquinol is oxidised with the transfer of electrons to complex I
d. Oxygen acts as terminal electron acceptor

(1) a, b and d only

(2) a, c and d only

(3) b and d only

CC-716 (4) b and c only CC-716

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105. Choose the **correct** match

(1) Alcoholic fermentation – Six CO_2 molecules are formed from one glucose molecule

(2) Aerobic respiration – Six H_2O molecules are formed from one glucose molecule

(3) Respiratory quotient – Is ratio of CO_2 consumption to O_2 evolution

(4) Glycolysis – Leads to formation of PEP as end product

(1) (1)

(2) (2)

(3) (3)

(4) (4)

106. Choose the option which is **true** for both lactic acid and alcoholic fermentation

(a) Less than 7% of the energy is released from glucose

(b) CO_2 is released

(c) All the released energy from glucose gets trapped as high energy bonds of NADPH

(d) Net gain is 2ATP

CC-716

In the light of above statements, select the **correct** answer from the given options.

(1) Only a is correct

(2) Only a and d are correct

(3) Only b and c are correct

(4) Only b and d are correct

107. The steps of Krebs cycle where decarboxylation occurs are

(a) Pyruvic acid → Acetyl CoA

(b) Oxalosuccinic acid → α-KGA

(c) Succinyl-CoA → Succinic acid

(d) α-KGA → Succinyl CoA

The correct ones are

(1) (a), (b) and (d) only

(2) (b), (c) and (d) only

(3) (b) and (d) only

(4) All (a), (b), (c) and (d)

108. How many of the following statements is/are **incorrect**?

a. Pyruvate undergoes oxidative decarboxylation by complex set of reactions after it enters mitochondrial matrix.

b. Conversion of succinyl CoA to succinic acid is an example of oxidative phosphorylation.

c. Oxygen drives the whole process of oxidative phosphorylation by removing hydrogen from the system.

CC-716

d. In both fermentation and aerobic respiration only partial breakdown of glucose occur to CO_2 and H_2O.

(1) One

(2) Two

(3) Three

(4) Four

109. The first member of citric acid cycle is

(1) Oxaloacetic acid

(2) Citric acid

(3) Acetyl CoA

(4) Pyruvic acid

110. During the conversion of 3-phosphoglyceraldehyde to 1, 3-bisphosphoglycerate in glycolysis

(1) Energy of ATP is used

(2) Two molecules of water is released

(3) ADP is converted into ATP

(4) $NADH + H^+$ is formed

111. Which of the given metabolites is common in respiration mediated breakdown of fat and protein?

(1) Dihydroxyacetone phosphate

(2) Glyceraldehyde 3-phosphate

(3) Acetyl CoA

CC-716

(4) Phosphoenol pyruvate

CC-716

112. All of the following are related to complex IV of mitochondria, **except**

(1) Cytochrome a and a_3

(2) Cytochrome c oxidase

(3) Two copper centers

(4) Cytochrome c

113. Which plant growth regulator was isolated from autoclaved herring sperm DNA?

(1) Auxins

(2) Gibberellins

(3) Cytokinin

(4) Ethylene

114. If the apical bud is removed from a plant then it will lead to

(1) Yellowing of leaves

(2) Decline in growth of lateral branches

(3) Inhibition of flowering

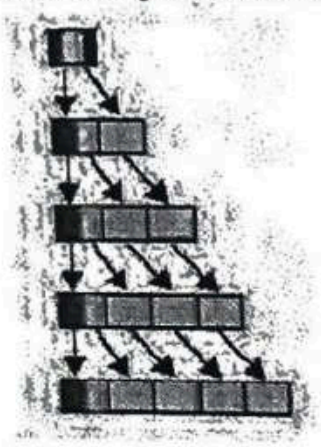
(4) Activation of axillary buds

CC-716

CC-716

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115. Select the **incorrect** statement w.r.t. given diagram in which the boxes with dark shade show cells capable of division and cells with light shade show cells that lose capacity to divide.



(1) This type of growth is shown by all the cells, tissues and organs of a mature flowering plant

(2) In this type of growth only one daughter cell continues to divide while the other differentiate and mature

(3) This type of growth is exemplified by a root elongating at constant rate

(4) It gives linear curve when growth is plotted against time

CC-716

116. Read the following assertion (A) and reason (R) and select the **correct** option.
Assertion (A): In juvenile conifers, gibberellins has been found to cause quicker early growth so that maturity is reached early.
Reason (R): Gibberellins is useful for obtaining early and quicker yield of economically important seeds.

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is the not correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false.

117. Leaf 'A' of initial size 10 cm^2 grows 5 cm^2 per unit time and leaf 'B' of 25 cm^2 grows 5 cm^2 per unit time.
The relative growth rates of leaves A and B respectively are

(1) 20% and 25%

(2) 10% and 20%

(3) 50% and 20%

(4) 50% and 25%

118. In order to increase the yield of pineapple plants, flowering is paramount. But pineapple plants take very long time to flower. Which of the following combination of plant hormones should be applied on pineapple plants by the farmer to induce flowering?

(1) GA and cytokinin

(2) Auxin and ethylene

(3) Cytokinin and ABA

(4) GA and ABA

119. Read the following statements.
Statement A. Ethephon is used as aqueous solutions from which a volatile plant hormone is released slowly
Statement B. The gaseous hormone is used for artificial ripening of mango.
Select **correct** option w.r.t above A and B statements.

(1) Both statements A and B are true

(2) Only statement B is true

(3) Only statement A is true

(4) Both statements A and B are false

120. ABA is also called stress hormone because

(1) It increases the stress conditions for the plants

(2) It regulates the opening and closing of stomata

(3) It increases the tolerance of plants in adverse environmental conditions

(4) It increases the water absorption capacity of the plant

121. Which of the following plants exhibit heterophyllous development due to environment?

CC-716

(1) Buttercup

CC-716

(2) Cotton

(3) Coriander

(4) Larkspur

122. In the geometric growth equation, $W_1 = W_0 e^{rt}$, 'e' is

(1) Growth rate

(2) Time of growth

(3) Base of natural logarithm

(4) Initial size

123. Auxin was first isolated from

(1) Degraded product of autoclaved herring sperm DNA

(2) Human urine

(3) Rice seedlings

(4) Coconut milk

124. As an outbreeding device, self incompatibility is

(1) A genetically controlled mechanism

(2) The germination of pollen grains before its maturity

(3) The germination of pollen grains within the anther

CC-716

CC-716

(4) Failure of pollination due to lack of pollinating agents

125. The hilum

(1) Is a scar on the fruit where pedicel was present

(2) Is the region where body of the ovule fuses with funicle

(3) Encircle the nucellus completely

(4) Is the stalk that attaches the ovule with placenta

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126. Dehydration and dormancy of mature seeds are
(a) Crucial for storage of seeds
(b) Important for raising crop in the next season
(c) Is crucial for germination of seeds
In the light of above statements choose the **correct** answer from the options given below.

(1) (a) and (b) only

(2) (b) and (c) only

(3) (a) and (c) only

(4) Only (c)

127. In male gametophyte, (A) leads to 2-celled stage and shedding of pollen grain at this stage occurs in (B) angiosperm.
Fill in the above blanks by choosing the **correct** option for (A) and (B) respectively.

(1) Meiosis; 40%

(2) Asymmetric meiosis; 60%

(3) Asymmetric mitosis; 60%

(4) Mitosis; 40%

CC-716

128. Select the **incorrect** statement w.r.t. apomixis.

(1) Seen in a few flowering plants such as some species of Asteraceae and grasses.

(2) It is a special mechanism, to produce seeds without fertilisation.

(3) Segregation of characters is very easy in apomicts.

(4) Apomixis is a form of asexual reproduction that mimics sexual reproduction.

129. In *Vallisneria*

(1) Pollen grains are light and wettable

(2) Flowers possess well exposed stamens

(3) Pollen moves actively in water

(4) Pollination takes place on the surface of water

130. Which of the following option depicts the **correct** sequer of embryo sac formation?

(1) Microspore mother cell → Megaspores → embryo sac

(2) Megasporangium → Sporogenous tissue → Microspo mother cell → Embryo sac

(3) Megasporangium → megaspore mother cell → Megaspore → Embryo sac

(4) Ovule → Megaspore mother cell → Nucellus → Embryo sac

131. In *Michelia*

(1) Gynoecium is syncarpous with more than one carpels

(2) Gynoecium consists of single carpel

(3) Gynoecium is absent

(4) Gynoecium is apocarpous

132. Proximal end of the filament of stamen is attached to the

(1) Anther

(2) Placenta

(3) Connective

(4) Thalamus or petal

CC-715

133. Water pollination is observed in

(1) Grasses

(2) Water hyacinth

(3) Water lily

(4) Hydrilla

134. Vegetative cell of pollen grains

(1) Floats in the cytoplasm of generative cell

(2) Has large irregular shaped nucleus

(3) Produces male gametes by mitosis

(4) Is smaller than generative cell

135. Endosperm is not present in the mature seeds of all, **except**

(1) Pea

(2) Gram

(3) Castor

(4) Bean

CC-716 CC-716 CC-716

Aakash Medical|IIT-JEE|Foundations

ZOOLOGY

136. Select the correct statement.

(1) Body form of adult *Aurelia* is polyp while larval form is medusa.

(2) *Meandrina* has an endoskeleton made up of spicules or spongin fibres.

(3) In *Obelia*, polyps produce medusae asexually.

(4) *Pennatula* exhibits only cellular level of body organisation.

137. Consider the given statements.

Statement (A): All the acoelomate animals are radially symmetrical while all pseudocoelomates are bilaterally symmetrical.

Statement (B): All animals with radial symmetry exhibit organ-system level of body organisation.

Select the correct option.

(1) Both statements (A) and (B) are correct

(2) Both statements (A) and (B) are incorrect

(3) Only statement (A) is correct

(4) Only statement (B) is correct

CC-716

138. Water vascular system is present in the members of phylum 'X'. Choose the correct statement w.r.t. the members of 'X'.

(1) Complete digestive system and advanced excretory system are present in them

(2) Adults are radially symmetrical in nature

(3) They are hermaphrodites with asexual mode of reproduction

(4) Development is direct and fertilisation is internal in them

139. Observe the figure given below



Select the incorrect option w.r.t it.

(1) It represents cnidoblast in its expelled stage

(2) Represents nematocysts which are present only on the tentacles of ctenophores

(3) This cell is used for defense in *Adamsia*.

(4) Represents the cell that contains stinging capsules in the *Gorgonia*

140. Select the **correct** match w.r.t. organism, its phylum and characteristic feature.

	Organism	Phylum	Characteristic feature
(1)	Liver fluke	Platyhelminthes	Have both hooks and suckers
(2)	Scorpion	Annelida	Body is drvided into head, thorax and abdomen
(3)	Cuttle fish	Mollusca	Monoecious and oviparous
(4)	Brain coral	Cnidaria	Gastrovascular cavity with a single opening

(1) (1)

(2) (2)

(3) (3)

(4) (4)

CC-716 CC-716

141. Read the following statement.

"Body is covered by calcareous shell and is unsegmented with a distinct head, muscular foot and visceral hump."

Choose the set of organisms from the options given below which is best described by the above mentioned statement.

(1) *Locusta, Hirudinaria*

(2) *Pila, Dentalium*

(3) *Asterias, Cucumaria*

(4) *Ascaris, Taenia*

142. Match the column I with column II.

	Column I		Column II
a.	Collar cells	(i)	Ctenophores
b.	Metagenesis	(ii)	Hemichordates
c.	Comb jellies	(iii)	Sponges
d.	Proboscis gland	(iv)	Coelenterates

Choose the **correct** option.

(1) a(i), b(ii), c(iii), d(iv)

(2) a(iii), b(ii), c(i), d(iv)

(3) a(iii), b(iv), c(i), d(ii)

(4) a(iii), b(i), c(ii), d(iv)

143. Consider the given features :

a. Absence of excretory system

b. Possesses complete digestive system

c. Difference of symmetry between adult and larval forms

d. Dioecious

Select the organism that possesses all of the above given features.

(1) *Aplysia*

(2) *Antedon*

(3) *Ascaris*

(4) *Anopheles*

CC-716 CC-716

144. Which of the following set of animals have same level of body organisation?

(1) Sycon, *Hydra*, *Fasciola*

(2) *Ascaris*, *Pheretima*, *Pila*

(3) *Taenia*, *Ctenoplana*, *Nereis*

(4) *Wuchereria*, *Pheretima*, *Spongilla*

145. Given below are few features:
(i) Body is circular in cross-section
(ii) Three germ layers are present but body cavity is not completely lined by mesoderm
(iii) Lacks digestive system thus, absorbs nutrients from the host directly through body surface
(iv) Have similar level of body organisation as *Nereis*
How many of the above given features is/are correct for *Ancylostoma*?

(1) Four

(2) Three

(3) Two

(4) One

CC-716

146. In echinoderms, digestive system is complete with mouth on _____ side and anus on the _____ side.
Select the **correct** option that fills the respective blanks.

(1) Upper; lower

(2) Ventral; dorsal

(3) Dorsal; lateral

(4) Dorsal; ventral

147. Which of the following options does **correctly** represent the characteristic features of members of phylum to which *Nereis* belongs to?

(1) Triploblastic, unsegmented body and bilaterally symmetrical

(2) Triploblastic, circular body and radially symmetrical

(3) Triploblastic, segmented body and bilaterally symmetrical

(4) Triploblastic, mostly parasitic and acoelomate condition

148. **Assertion (A)** : During dissection and study of an adult *Hirudinaria*, both ovaries and testes were found in the same segmented worm.
Reason (R): Leeches are dioecious and perform sexual reproduction.
In the light of above statements, choose the **correct** option

CC-716

(1) Both (A) and (R) are true and (R) is the correct explanation of (A)

(2) Both (A) and (R) are true but (R) is not the correct explanation of (A)

(3) (A) is true but (R) is false

(4) Both (A) and (R) are false

149. In sponges, fertilisation is A and development is
Choose an option which fills the blanks A and B correc

(1) A - External, B - Indirect

(2) A - External, B - Direct

(3) A - Internal, B - Indirect

(4) A - Internal, B - Direct

150. Which of the following features is common betw
Balanoglossus and *Echinus*?

(1) Body is composed of proboscis, collar and trunk

(2) Only organ level of body organisation

(3) Resides exclusively in marine habitat

(4) Closed circulatory system

151. Given below are two statements: one is labelled as **Assertion (A)** and the other is labelled as **Reason (R)**.
Assertion (A) : All vertebrates are chordates but all chordates are not vertebrate.
Reason (R) : The members of subphylum vertebrata possess notochord during the embryonic period, the notochord is replaced by cartilaginous or bony vertebral column in adults.
In the light of the above statements, choose the **correct** answer from the options given below:

CC-716 CC-716

(1) **(A)** is true but **(R)** is false

(2) **(A)** is false but **(R)** is true

(3) Both **(A)** and **(R)** are true and **(R)** is the correct explanation of **(A)**

(4) Both **(A)** and **(R)** are true but **(R)** is **not** the correct explanation of **(A)**

152. Read the following statements:
Statement A: All tetrapods are gnathostomes but all gnathostomes are not tetrapods.
Statement B: All gnathostomes are vertebrates but all vertebrates are not gnathostomes.
Choose the **correct** option.

(1) Both statements A and B are correct

(2) Both statements A and B are incorrect

(3) Only statement A is correct

(4) Only statement B is correct

153. Given below is a list of features:
(i) Ear is represented by tympanum.
(ii) Oviparity is seen.
(iii) Eyes have eyelids.
(iv) Skin is moist without scales.

CC-716 CC-716

The above given features are associated with members of class

(1) Osteichthyes

(2) Chondrichthyes

(3) Amphibia

(4) Reptilia

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154. Animal 'J' belongs to the same class as snakes. It has two auricles and two ventricles in its heart.
Identify the animal from the following and select the correct option.

(1) Bat

(2) Crocodilus

(3) Chameleon

(4) Necturon

155. Members of the class Chondrichthyes and Osteichthyes differ from each other with all of the following features, except

(1) Streamlined body

(2) Gill cover

(3) Claspers in the pelvic fins of males

(4) Placoid scales

156. How many animals given in the box below is/are not able to regulate their body temperature?

Ichthyophis, Bufo, Rana, Chameleon, Calotes, Anemidyes, Pteropus, Macropus

Select the correct option.

(1) One

(2) Two

(3) Five

(4) Four

157. The scaly hindlimbs of birds are modified for performing all of the given functions, except

(1) Walking

(2) Flying

(3) Swimming

(4) Clasping the tree branches

158. Select the option which represents the correct set of differences between bony fishes and cartilaginous fishes.

	Bony fishes	Cartilaginous fishes
(1)	Mouth is always located ventrally	Mouth is terminal
(2)	Skin contains placoid scales	Skin is covered with ctenoid scales
(3)	Claspers are absent in males	Males pelvic fins bear claspers
(4)	Fertilisation is usually internal	They exhibit external fertilisation

(1) (1)

(2) (2)

(3) (3)

(4) (4)

159. In members of the phylum Chordata, notochord is present 'X' in gut but 'Y' in nerve cord.

Select the correct option to fill in the respective blanks.

(1) X - Ventral; Y - Dorsal

(2) X - Lateral; Y - Dorsal

(3) X - Dorsal; Y - Ventral

(4) X - Ventral; Y - Lateral

160. Select the option representing only the correct statements w.r.t. morphology of frogs.
(a). A membranous tympanum is present on either side of their eyes.
(b). They do not have webbed digits in hind limbs for swimming.
(c). Mucous glands are present in their highly vascularised skin.
(d). Their body is divisible into head, neck and trunk.

(1) (a) and (c)

(2) (b) and (d)

(3) (a) and (b)

(4) (c) and (d)

161. Which among the following organisms exhibits internal fertilisation, oviparity as well as possesses hair on its skin?

(1) Ichthyophis

(2) Pteropus

(3) Ornithorhynchus

(4) Necturon

162. Crocodilus and Macropus differ from Chameleon and Calotes as former have/show

(1) Pulmonary respiration

(2) Notochord during the embryonic stage

(3) Direct development

(4) Four chambered heart

163. Animals which are devoid of scales and paired fins and whose adults are marine while larvae are fresh water forms, belong to one of the taxon under

(1) Division Gnathostomata

(2) Super class Pisces

(3) Class Amphibia

(4) Subphylum Vertebrata

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164. The route for passage of sperms in frog is given below.
Testis → (A) → Kidneys → (B) → Urinogenital duct → (C).
Identify (A), (B) and (C) and choose the correct option.

	(A)	(B)	(C)
(1)	Vas deferens	Bidder's canal	Ureter
(2)	Vasa efferentia	Vas deferens	Cloaca
(3)	Vasa efferentia	Bidder's canal	Cloaca
(4)	Epididymis	Nephron	Cloacal aperture

(1) (1)

(2) (2)

(3) (3)

(4) (4)

165. Choose the correct option to complete the analogy w.r.t. respiration in frog.
Cutaneous : Skin : : _____ : Lungs

(1) Branchial

(2) Cutaneous

(3) Pulmonary

(4) Buccal

CC-716

166. The special venous connection present between kidneys and lower parts of the body is present in

(1) Frogs called renal portal system

(2) Humans called renal portal system

(3) Frogs called hepatic portal system

(4) Humans called hepatic portal system

167. Choose the highly vascularised structure in adult frog which helps in respiration in water as well as on land.

(1) Gills

(2) Skin

(3) Buccal cavity

(4) Lungs

168. Comprehend the following w.r.t. the nervous system of frogs.
(a) Hind brain consists of cerebellum, pons and medulla oblongata
(b) 20 cranial nerves arise from brain
(c) Brain is enclosed in a cartilaginous structure called the brain box.
Choose the option that represents the incorrect set of features only.

(1) (a), (b) and (c)

(2) (a) and (c)

(3) (a) and (b)

(4) (b) and (c)

169. In frogs, the sense organ which is a cellular aggregation around the nerve endings is organ of

(1) Touch

(2) Vision

(3) Balance

(4) Hearing

170. The forelimbs of male frog can be differentiated from its hind limbs on the basis of

(1) Swimming

(2) Leaping

(3) Presence of copulatory pad

(4) Burrowing

171. The alimentary canal of a frog is *(A)* because frog is a carnivore and hence the length of intestine is *(B)*.
Choose the option that **correctly** identifies (A) and (B) respectively.

(1) Short, reduced

(2) Long, elongated

(3) Short, elongated

(4) Long, reduced

172. The urinary bladder in a female frog is located

(1) Lateral to the rectum

(2) Dorsal to the rectum

(3) Ventral to the rectum

(4) Anterior to the rectum

173. Titillator and pseudopenis are associated with which of the following structures of a cockroach?

(1) Left phallomere

(2) Right phallomere

(3) Ventral phallomere

(4) Long tubules

174. Choose the incorrect statement w.r.t. the structure present in cockroach's alimentary canal that helps in grinding the food particles.

(1) It is also known as proventriculus.

(2) It has an outer layer of thick circular muscles.

(3) It has thick inner cuticle forming six highly chitinous plate

(4) It is followed by a sac-like structure called gizzard.

Aakash Medical | IIT-JEE | Foundations
CC-716 CC-716 CC-716 CC-716 CC-716

175. In a male *Periplaneta americana*, the mushroom shaped gland is present in _____ abdominal segments.
Choose the correct option.

(1) 6th - 7th

(2) 9th - 10th

(3) 4th - 6th

(4) 2nd - 6th

176. Select the **incorrect** statement w.r.t *Periplaneta americana*.

(1) On an average, females produce 9-10 oothecae, each containing 14-16 eggs.

(2) Development is paurometabolous.

(3) Nymph grows by moulting about 13 times.

(4) Next to last nymphal stage has fully developed wings.

177. Cockroach is 'X' whereas an adult frog is 'Y' w.r.t the main nitrogenous waste products.
Select the correct option w.r.t 'X' and 'Y'.

(1) X - Ureotelic; Y - Uricotelic

(2) X - Uricotelic; Y - Ureotelic

(3) X - Ammonotelic; Y - Uricotelic

(4) X - Ureotelic; Y - Ammonotelic

178. Read the following statements.

a. Nervous system of cockroach consists of a series of segmentally fused ganglia located on dorsal side of body

b. In cockroach, sensory structures are absent on maxillary and labial palps.

c. Supra-oesophageal ganglion functions as brain in cockroach

d. Mosaic vision in cockroach has less sensitivity but resolution is high

Select the option representing only the correct statements .

(1) a and b

(2) c and d

(3) Only c

(4) a,b,c and d

179. The mouthpart of cockroach that possesses grinding and incising region is 'X'. The number of 'X' is equal to the number of

(1) Labium

(2) Maxilla

(3) Proventriculus

(4) Ostia

180. How many of the structures given in the box below are associated with the 6th abdominal segment of female cockroaches?

Mushroom-shaped gland, Boat-shaped sclerite, Spermathecal, Testes, Ovaries, Alary muscles

Select the **correct** option.

(1) Three

(2) Four

(3) Two

(4) Five

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CC-716 CC-716 CC-716 CC-716 CC-716